

## **Program Courses Outline**

The program consists of two tracks each aimed at a specific career path:

1. Data Scientist
2. AI Engineer

### **Pre-requisites**

Both tracks presume the following about the learner to get started:

1. English B2 (Upper-Intermediate) level: IELTS 6.5 or TOEFL 80.
2. Algorithmic thinking and problem-solving skills.
3. Strong foundation in programming.
4. Working laptop with internet access.

## **Track 1: Data Scientist**

Time Estimate: 4-5 weeks at 30 hrs/week.

### **1. Data Wrangling**

Fundamentals of data wrangling and analysis in Python via pandas, matplotlib and seaborn.

- M1. Filtering, Sorting, and Aggregation
- M2. Data Wrangling
- M3. Data Visualization
- M4. Timeseries Analysis

Time Estimate: 5 days x 6 hours.

### **2. Data Science**

Calculate, analyze, visualize, and extract insights from data. Formulate hypotheses and draw conclusions.

- M1. Introductions
- M2. Univariate Analysis
- M3. Bivariate Analysis
- M4. Inferential Statistics

Time Estimate: 5 days x 6 hours.

### **3. Applied Machine Learning**

Build reliable predictive modeling pipelines, debug its issues, evaluate and compare alternatives.

- M1. Supervised ML: Regression and Classification
- M2. Estimating and Improving Model Generalization Performance
- M3. Pipeline: Building Reliable Predictive Models
- M4. Decision Trees and Ensembles
- M5. AutoML

Time Estimate: 10 days x 6 hours.

## **Track 2: AI Engineer**

Time Estimate: 4-5 weeks at 30 hrs/week.

### **1. Agentic Engineering**

Work effectively and efficiently with AI in software engineering projects.

- M1. From Vibe Coding to Agentic Engineering
- M2. Skills for Engineers
- M3. Agent Modes

Time Estimate: 5 days x 6 hours.

### **2. Building Agentic AI Software**

Develop, debug, evaluate, deploy, and monitor LLM-driven AI Agents to automate tasks involving unstructured data.

- M1. Signatures and Modules
- M2. Agents with Tools
- M3. Coding Agents
- M4. Optimization
- M5. Retrieval Augmented Generation (RAG)

Time Estimate: 10 days x 6 hours.

### **3. Applied Deep Learning**

Select, use, compose, fine-tune, and deploy open-weight deep learning models on various unstructured data tasks.

- M1. HuggingFace and Large Language Models
- M2. Realtime Computer Vision Models

Time Estimate: 5 days x 6 hours.